

DeepClean AI

Automated Underwater Marine Debris & Seabed Anomaly Detection Web Platform

Category: Marine Technology & Environmental AI

Modality: Side-Scan Sonar (Acoustic Imagery)

Document Type: System Architecture & PRD

Target Agencies: Coast Guard, NIOT, Port Authorities

Primary Stack: FastAPI, YOLOv8/11, Next.js, PostGIS

Status: Ready for Prototype Development

1. Executive Summary & Problem Scope

Marine pollution poses severe ecological and maritime hazards. Millions of tons of synthetic fishing gear (*ghost nets*), plastic containers, sunken drums, and wreckage litter the ocean floor. At water depths exceeding 20 meters, extreme turbidity, suspended sediment, and total lack of sunlight render optical cameras ineffective. Consequently, **Side-Scan Sonar (SSS)** utilizing acoustic backscatter is the standard for hydrographic seafloor mapping.

However, conventional hydrographic workflows require marine geophysicists to manually analyze hundreds of continuous gigabytes of acoustic waterfall plots. This manual inspection causes a severe backlog, high analytical costs, and delayed salvage operations. **DeepClean AI** provides a web-based, end-to-end intelligence suite that ingests raw acoustic survey files, filters speckle reverberation, executes real-time AI inference for debris classification, and generates actionable geospatial salvage dossiers for response teams.

THE ACOUSTIC VISION PHYSICS

Sonar images do not depict visual color; they record acoustic return intensity. An underwater object is defined by two complementary components: **(1) Acoustic Highlight** (high backscatter energy reflection) followed immediately by **(2) Acoustic Shadow** (pitch-black zone devoid of sound reflections). Object height above the seabed is geometrically calculated as: $H_{obj} = (L_{shadow} \times H_{sensor}) / R_{range}$.

2. End-to-End System Architecture

The platform adopts a decoupled microservice architecture ensuring low latency during high-resolution hydrographic tile inference and georeferenced spatial retrieval.

```
[Raw Survey Ingestion] --> [.XTF / .JSF Sonar Logs, GeoTIFF, 16-bit PNG] | ▼ [Pre-Processing Engine] --> [pyxtf parser + Lee Speckle Filter + CLAHE Equalization] | ▼ [Deep Learning Engine] --> [Custom YOLOv8/YOLO11-Sonar + Acoustic Shadow Segmentation] | ▼ [Spatial Backend] --> [FastAPI Asynchronous Gateway + PostgreSQL / PostGIS DB] | ▼ [Web Client & GIS UI] --> [Next.js + Tailwind CSS + Mapbox GL JS / Leaflet Engine] | ▼ [Dossier Engine] --> [Automated PDF Salvage Mission Dossier Generator]
```

3. Technology Stack Specification

Layer	Technology	Role & Responsibility
Frontend Interface	Next.js 14, Tailwind CSS, Lucide	Responsive web portal, responsive waterfall viewer, telemetry HUD.

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GIS & Mapping	Mapbox GL JS, Leaflet, Turf.js	Geospatial visualization of survey tracks, debris cluster heatmaps.
Backend API	Python FastAPI, Uvicorn, Celery	High-throughput asynchronous REST API for model serving and uploads.
Computer Vision	PyTorch, Ultralytics YOLOv8/11, OpenCV	Acoustic highlight-shadow pair detection, bounding box regression.
Hydrographic Processing	pyxtf, Rasterio, GDAL	Parsing binary eXtended Triton Format (.XTF) pings & Slant-Range Correction.
Database & Spatial Engine	PostgreSQL 16 + PostGIS extension	Spatial indexing (R-Tree), debris coordinates, mission registry.

4. Core Functional Modules of the Web Platform

1. Survey Ingestion & Waterfall Player

Supports drag-and-drop uploads of `.XTF`, `.JSF`, and GeoTIFF files. Features an HTML5 Canvas acoustic waterfall player simulating real-time towfish sweeps with interactive gamma and contrast (CLAHE) sliders.

2. AI Automated Detection Studio

Executes inference in $<120\text{ms}$ per sonar tile. Displays localized bounding boxes, target classification, confidence percentage, and automated acoustic shadow length estimation.

3. Interactive Maritime GIS Map

Overlays surveyed ship tracks onto satellite bathymetric charts. Categorizes targets into color-coded risk markers (Red: Ghost Nets, Amber: Metal Drums, Blue: Debris).

4. Salvage Dossier Export Engine

Generates standardized operational PDF reports with survey duration, total seafloor area swept (km^2), precise GPS salvage waypoints, and cropped sonar evidence snippets.

5. Target Classification Schema & Training Data

The detection network is optimized for five primary object classes encountered in coastal and shelf waters:

Target Class	Hazard Level	Acoustic Characteristics	Primary Data Source
Ghost Fishing Net	Critical Ecological	Amorphous, fibrous mesh texture, uneven diffuse acoustic shadow.	Marine Debris Sonar Dataset
Submerged Metal Drum	Toxic Hazard	Cylindrical strong acoustic return, distinct rectangular shadow.	AquaScan-1K / Synthetic SSS
Discarded Tyre	Marine Waste	Toroidal circular highlight, hollow shadow contour.	Forward-Looking Sonar Debris
Wreckage Fragment	Navigational	Sharp angular edges, high-contrast backscatter signature.	SeabedObjects-KLSG Benchmark
Underwater Pipeline	Infrastructure	Continuous linear feature with parallel uninterrupted shadow.	OpenBathy NOAA Hydrographic

6. Database Schema (PostgreSQL + PostGIS)

```
-- Table: Acoustic Surveys CREATE TABLE surveys ( id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
survey_name VARCHAR(120) NOT NULL, vessel_name VARCHAR(100), file_format VARCHAR(10) CHECK (file_format
IN ('XTF', 'JSF', 'TIFF', 'PNG')), swath_width_meters FLOAT DEFAULT 150.0, area_covered_sqkm FLOAT,
track_geom GEOMETRY(LineString, 4326), created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW() ); -- Table:
Detected Marine Debris Targets CREATE TABLE detected_debris ( id UUID PRIMARY KEY DEFAULT
gen_random_uuid(), survey_id UUID REFERENCES surveys(id) ON DELETE CASCADE, target_class VARCHAR(50)
NOT NULL, confidence_score FLOAT CHECK (confidence_score >= 0.0 AND confidence_score <= 1.0),
estimated_height_m FLOAT, location_geom GEOMETRY(Point, 4326) NOT NULL, crop_image_url TEXT,
salvage_status VARCHAR(20) DEFAULT 'PENDING' CHECK (salvage_status IN ('PENDING', 'ASSIGNED',
'RETRIEVED')), detected_at TIMESTAMP WITH TIME ZONE DEFAULT NOW() );
```

7. Page-by-Page Web UI Walkthrough

- Executive Dashboard** (`/`): High-level operational cards detailing total square nautical miles surveyed, tally of verified ghost nets, active salvage missions, and model accuracy metrics.
- Survey Live Studio** (`/survey/analyze`): Split-screen layout. Left pane plays the waterfall acoustic stream; right pane presents real-time YOLO bounding boxes, class probabilities, and manual annotation correction tools.
- Seabed GIS Map** (`/map`): Interactive marine geospatial console with toggleable layers for sonar tracks, debris heatmaps, bathymetric contours, and marine sanctuary boundaries.
- Debris Inventory & Registry** (`/targets`): Searchable, tabular registry of all identified underwater items with coordinates, status filters, and one-click coordinate export to navigation systems (GPX/CSV).

8. SIH Implementation Roadmap & Milestones

Phase	Core Deliverables	Evaluation Milestone
Phase 1: Week 1	Data curation, format parsing via pyxrf, dataset split (80/10/10).	Clean labeled baseline of 1,500+ sonar highlight-shadow patches.
Phase 2: Week 2	YOLOv8/11 fine-tuning, CLAHE augmentation, false-positive filtering.	Model achieves mAP@50 > 86% across debris classes.
Phase 3: Week 3	FastAPI backend, PostGIS spatial queries, Next.js frontend UI.	Working end-to-end prototype: upload file → get detections.
Phase 4: Week 4	Mapbox GIS integration, Canvas waterfall viewer, PDF export.	Live hackathon demonstration with real .XTF sonar files.